

IBRAHIM ADAM

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EDUCATION

California Institute of Technology

Sophomore, B.S. in Computer Science

GPA: 3.9/4.0

Pasadena, CA

Sep. 2025 – Jun. 2029

RELEVANT COURSEWORK

- Introduction to Computer Programming, Data Structures and Algorithms, Calculus of One and Several Variables
- Linear Algebra, Differential Equations

TECHNICAL SKILLS

Languages: Java, C, Go, Python, TypeScript, JavaScript, SQL, HTML/CSS

Tools/Frameworks: Spring Boot, React, Gin, Vite, Git, GitHub Actions, Docker Compose, PostgreSQL, Linux, Nginx

Concepts: REST APIs, Distributed Systems, Systems Programming, Concurrency, Authentication, Testing, CI/CD

EXPERIENCE

Software Engineer | TorMaMe

Jan. 2025 – Present

Remote

- Engineered **Python** backend services and **React/TypeScript** frontend components for TorMaMe's web platform, improving feature delivery speed by 40%.
- Built **REST APIs**, authentication flows, frontend interfaces, and backend integrations, reducing deployment failures and critical bugs by 30%.
- Automated deployment workflows using **Docker Compose**, **PostgreSQL**, and **Nginx**, cutting deployment time from 30 minutes to 10 minutes.

Software Engineering Intern | Veritas Foundation Inc.

Apr. 2025 – Present

Ashanti Region, Ghana / Remote

- Developed a digital outreach platform using **React** frontend components and backend **REST APIs** to coordinate school partnerships and student engagement.
- Implemented engagement tracking, partnership coordination workflows, and user data management features, increasing coordination efficiency by 45%.
- Translated outreach-team needs into scalable technical features supporting 20+ schools, sponsors, and community partners.

PROJECTS

Cross-Border Payments Simulator | Go, Gin, React, TypeScript, PostgreSQL, Docker Compose

Jan. 2026 – Mar. 2026

- Architected a distributed payments simulator modeling transfer initiation, FX quoting, route selection, ledger updates, and settlement tracking.
- Developed backend **REST APIs** in **Go/Gin** to simulate financial messaging, routing decisions, and transaction state transitions.
- Containerized services with **Docker Compose** and built **React** dashboards to visualize transaction states, reducing payment-flow debugging time by 50%.

OrderFlow | Java, Spring Boot, PostgreSQL, JUnit, GitHub Actions

Mar. 2026 – Apr. 2026

- Engineered a **Java/Spring Boot** trading engine supporting limit order submission, price-time-priority matching, trade execution, cancellation, and **REST APIs**.
- Implemented an in-memory matching engine with buy/sell priority queues, full and partial fills, and order status transitions, reducing debugging time by 60%.
- Added **JUnit** tests and **GitHub Actions** CI to validate builds on each push, improving regression detection and release confidence.

Hosty | C, Linux, Nginx, Systems Design

Oct. 2025 – Dec. 2025

- Built a multi-tenant static hosting platform in **C** with virtual hosting, authenticated APIs, and tenant-isolated routing.
- Implemented a custom HTTP server with thread pools, static file serving, path validation, rate limiting, and quota enforcement.
- Designed a persistent key-value store with append-only logging, in-memory indexing, and crash recovery, reducing recovery overhead by 35%.

LEADERSHIP / EXTRACURRICULAR

Rise Tutor

2025 – Present

- Tutored middle and high school students in math and science, improving average student performance from 45% to 78%.

Caltech Black Students Union, NSBE, and ColorStack

2025 – Present

- Active member of technical and identity-based communities focused on peer mentorship, professional development, and software engineering growth.